LIFTING SMOOTH AFFINE CHARTS THROUGH FIRST-ORDER THICKENINGS (EMERTON)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1497-1513 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1515-1551, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 28 of queue_stacks-stacks-more-morphisms.json (pb=1930, stmt 1497-1513, proof 1515-1551).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Editorial note. Statement carries a reference/slogan/history environment (patchenv -> patch-sd-envs.tex).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition lemma-gerbe-of-lifts: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1306–1309. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition lemma-gerbe-of-lifts-first-order: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1323–1333. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 06AD: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1551–1551. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition remark-gerbe-of-lifts: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 875–903. Cited by the

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0CK0: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1551–1551. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 072B: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1551–1551. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 05ZR: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1551–1551. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1 (Emerton)**.** *Email of Matthew Emerton dated April 27, 2016. Let $\mathcal{X} \subset \mathcal{X}'$ be a first order thickening of algebraic stacks. Let W be an affine scheme and let $W \to \mathcal{X}$ be a smooth morphism. Then there exists a cartesian diagram*

$$\begin{array}{ccc} W & \longrightarrow & W' \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{X}' \end{array}$$

with $W' \to \mathcal{X}'$ smooth and W' affine.

Proof. Consider the category $p : \mathcal{C} \to W_{spaces,étale}$ introduced in Remark remark-gerbe-of-lifts. The proposition states that there exists an object of $\mathcal{C}$ lying over W. Namely, if we have such an object $(W, W', a, i, y', \alpha)$ then $W = \mathcal{X} \times_{\mathcal{X}'} W'$. Hence $W \to W'$ is a thickening of algebraic spaces so W' is affine by More on Morphisms of Spaces, Lemma 05ZR and More on Morphisms, Lemma 06AD.

Lemma lemma-gerbe-of-lifts tells us $\mathcal{C}$ is a gerbe over $W_{spaces,étale}$. This means we can étale locally find a solution and these local solutions are étale locally isomorphic; this part does not require the assumption that the thickening is first order. By Lemma lemma-gerbe-of-lifts-first-order the automorphism sheaves of objects of our gerbe are abelian and fit together to form a quasi-coherent module $\mathcal{G}$ on $W_{spaces,étale}$. We will verify conditions (1) and (2) of Cohomology on Sites, Lemma 0CK0 to conclude the existence of an object of $\mathcal{C}$ lying over W. Condition (1) is true: the étale coverings $\{W_i \to W\}$ with each W_i affine are cofinal in the collection of all coverings. For such a covering W_i and $W_i \times_W W_j$ are affine and $H^1(W_i, \mathcal{G})$ and $H^1(W_i \times_W W_j, \mathcal{G})$ are zero: the cohomology of a quasi-coherent module over an affine algebraic space is zero for example by Cohomology of Spaces, Proposition 072B. Finally, condition (2) is that $H^2(W, \mathcal{G}) = 0$ for our quasi-coherent sheaf $\mathcal{G}$ which again follows from Cohomology of Spaces, Proposition 072B. This finishes the proof. □